

Estimación de la densidad

2025-08-21

Ingreso total familiar en CABA en el último trimestre de 2023

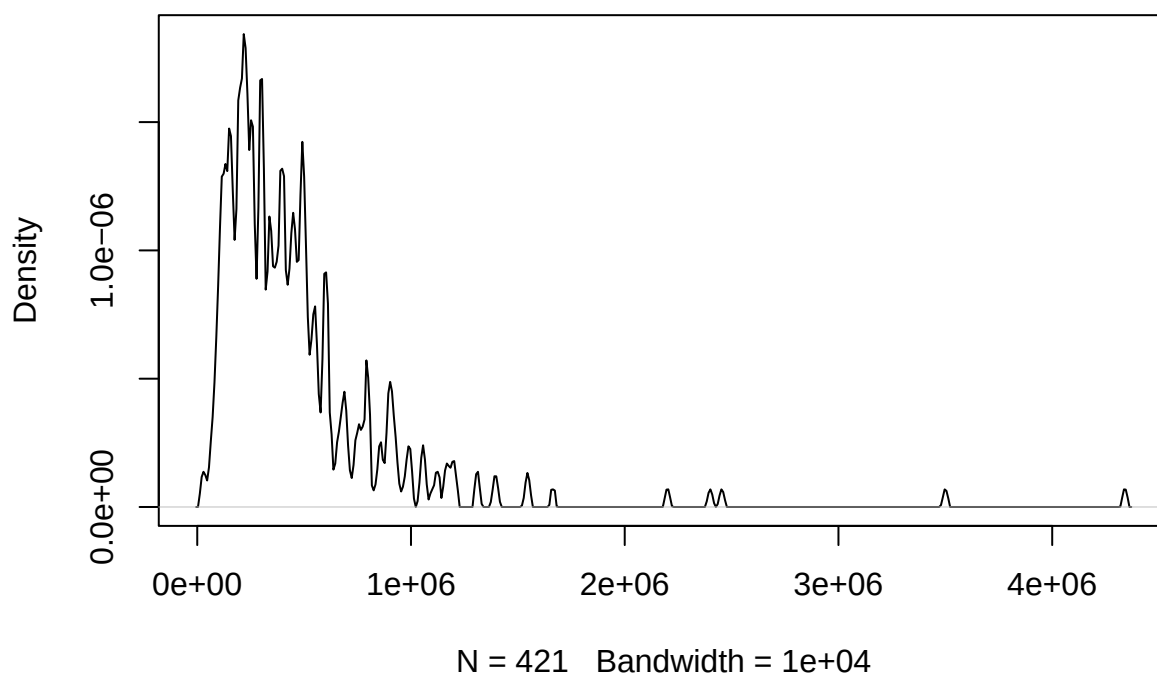
(Fuente: INDEC, [www.indec.gob.ar])

```
ITF <- read.csv("ITF2023_3.csv")[,2]
```

Núcleo rectangular

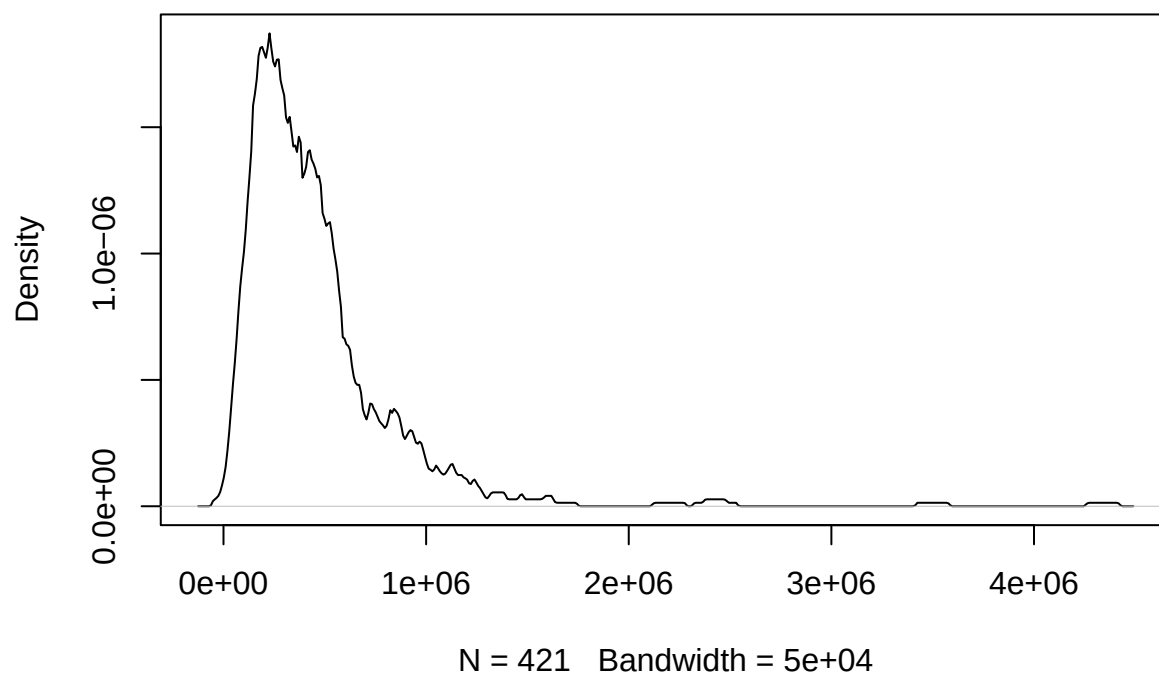
```
plot(density(ITF, bw=10000, kernel = "rectangular"))
```

density(x = ITF, bw = 10000, kernel = "rectangular")



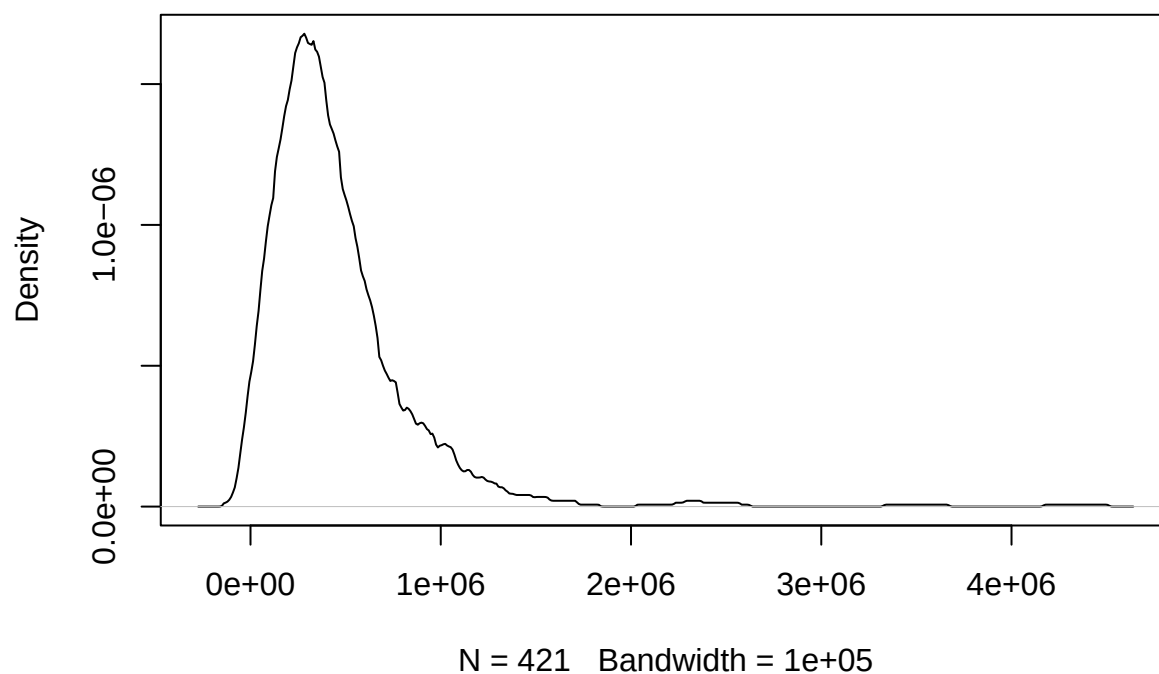
```
plot(density(ITF, bw=50000, kernel = "rectangular"))
```

density(x = ITF, bw = 50000, kernel = "rectangular")



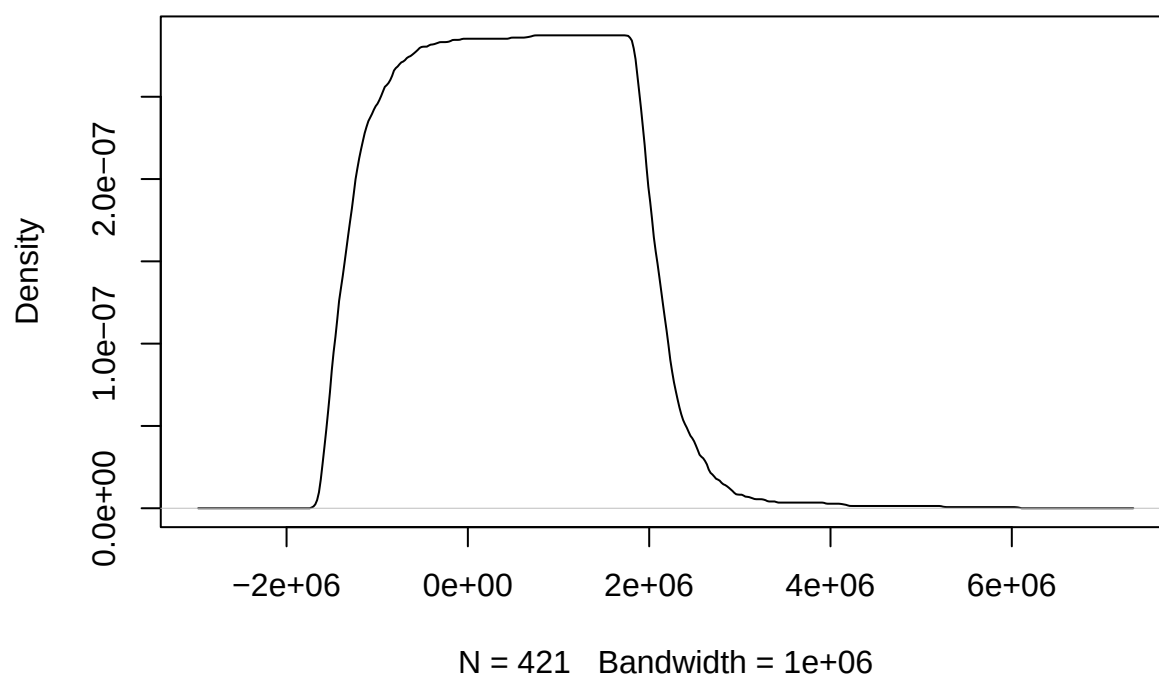
```
plot(density(ITF, bw=100000, kernel = "rectangular"))
```

density(x = ITF, bw = 1e+05, kernel = "rectangular")



```
plot(density(ITF, bw=1000000, kernel = "rectangular"))
```

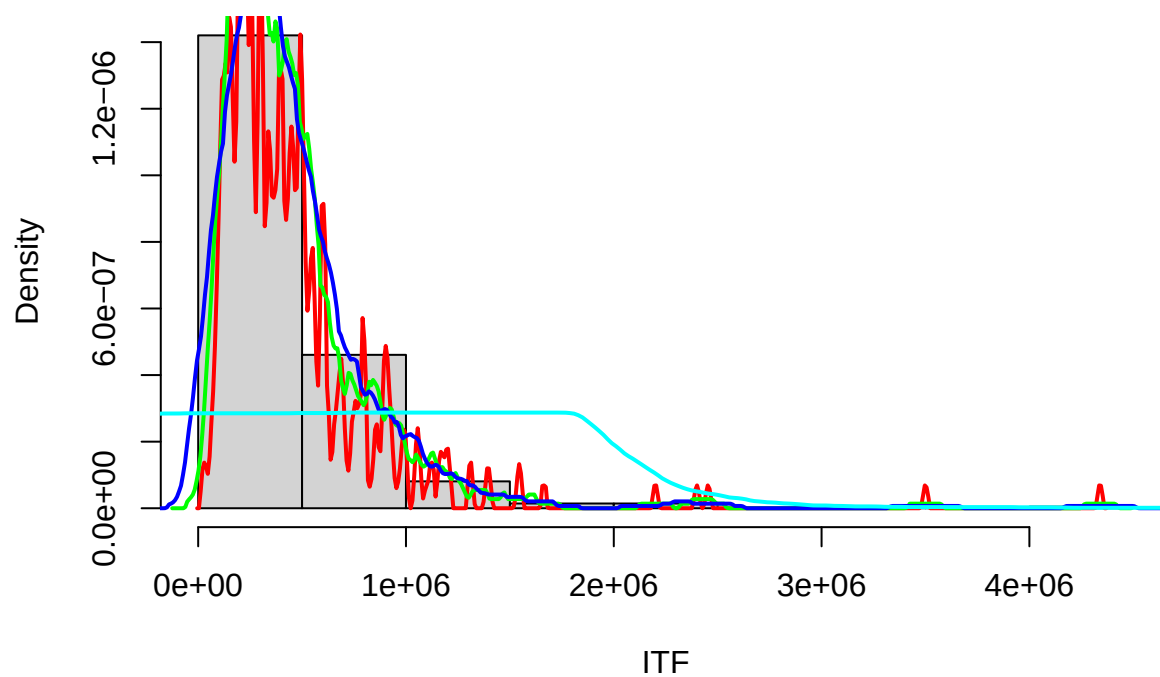
density(x = ITF, bw = 1e+06, kernel = "rectangular")



Comparación con el histograma

```
hist(ITF, probability = TRUE)
lines(density(ITF, bw=10000, kernel = "rectangular"), lwd =2, col = "red")
lines(density(ITF, bw=50000, kernel = "rectangular"), lwd =2, col = "green")
lines(density(ITF, bw=100000, kernel = "rectangular"), lwd =2, col = "blue")
lines(density(ITF, bw=1000000, kernel = "rectangular"), lwd =2, col = "cyan")
```

Histogram of ITF

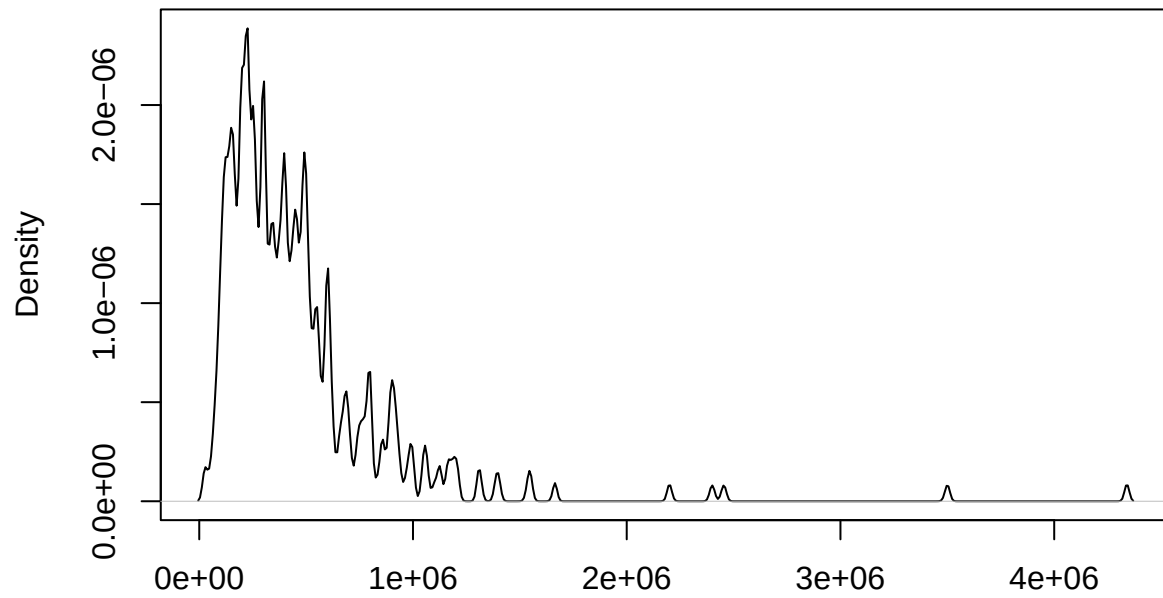


Núcleo gaussiano

Es el default en R

```
plot(density(ITF, bw=10000))
```

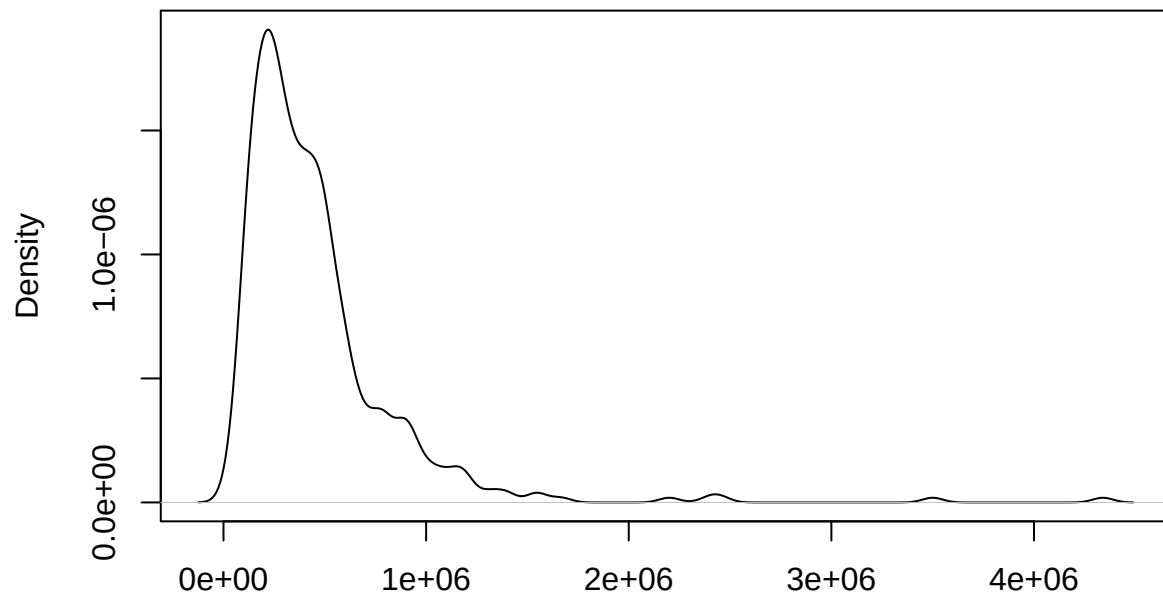
density(x = ITF, bw = 10000)



N = 421 Bandwidth = 1e+04

```
plot(density(ITF, bw=50000))
```

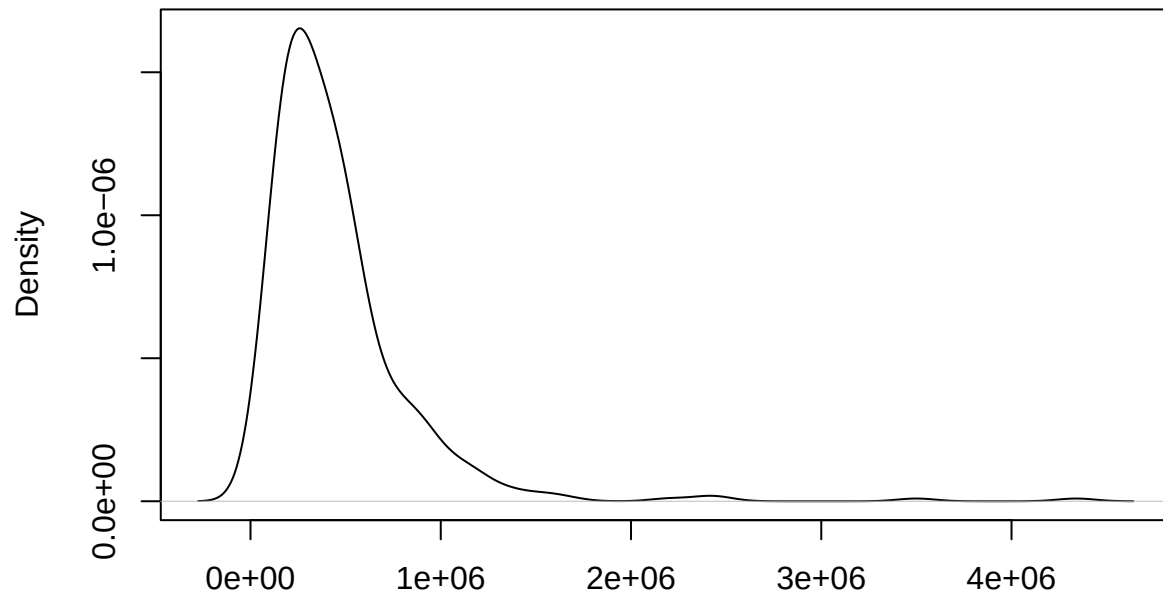
density(x = ITF, bw = 50000)



N = 421 Bandwidth = 5e+04

```
plot(density(ITF, bw=100000))
```

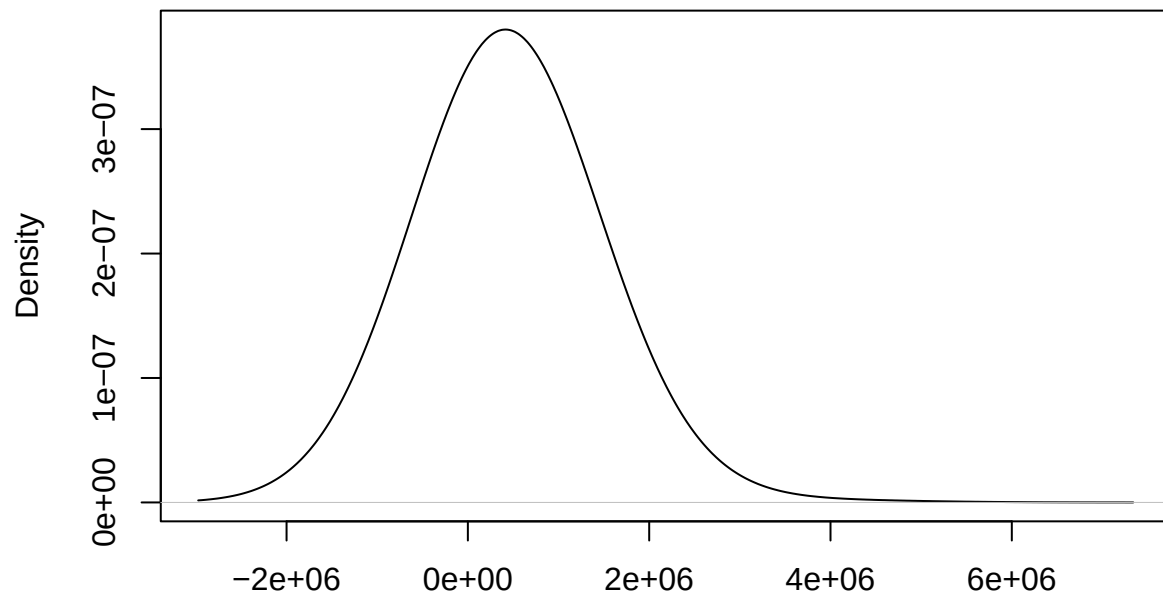
density(x = ITF, bw = 1e+05)



N = 421 Bandwidth = 1e+05

```
plot(density(ITF, bw=100000))
```

density(x = ITF, bw = 1e+06)



N = 421 Bandwidth = 1e+06